

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations
January 2023

20BT505 - BIOETHICS, BIOSAFETY AND IPR

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Elaborate on the ethical analysis framework with an example of bovine somatotrophin in dairy farming.	10	L*2	1	1,6,8
	b)	Write a descriptive note on socio-economic impacts of biotechnology.	10	L2	1	1,6,7
2.	a)	What are the ethical issues involved in stem cell research?	10	L2	1	6,8
	b)	Elucidate with examples, how microorganisms are used as bioweapons. Add a note on its ethical implications.	10	L2	1	6,8
		Unit – II				
3.	a)	Explain the term BSL. Briefly point out the various BSL and their significance in laboratory set up.	10	L2	2	1,6,8
	b)	What are the biosafety practices needed to be taken care in case of an institution?	10	L2	3	1,6,8
4.	a)	Elaborate on the various steps involved in risk management process.	10	L1	2	1,6,8
	b)	Give a brief note on how GLP was introduced. Add a note on the basic elements of GLP.	10	L2	3	1,6,8
5.	a)	What is Cartagena protocol? Add a note on its key elements.	10	L2	3	1,6,8
	b)	Illustrate on the biosafety regulations and guidelines with respect to genetic engineering techniques.	10	L2	3	1,6,8
		Unit – III				
6.	a)	Explain briefly the features of TRIPS agreement. Add a note on its basic requirements.	10	L2	4	6,8
	b)	Elaborate on the strategic goals and core activities of WIPO.	10	L2	4	6,8
7.	Write short notes on the following: i) Copyright ii) Budapest Treaty iii) Trade secret iv) Traditional knowledge		20	L2	4	6,8
8.	a)	Explain the steps involved in patenting	10	L2	5	6,8
	b)	Briefly explain the case of "Revocation of Turmeric Patent".	10	L2	5	6,8

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
